

T3 matched BSM $\Delta\mathcal{L}_{\text{EFT}}$ terms grouped by field content

EFT order: 5; loop order: 1. Columns are determined only from matched BSM EFT terms. Empty cells denote no matched term with that field combination.

Field combinations 1 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$H \ell$
									$+C_{H2}H_{a_1}H_{a_1}^\dagger$ $-H_{a_1}\mu^2H_{a_1}^\dagger$ $+\frac{\text{lambdaH1lnv1}^2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$ $+\frac{5\text{lambdaH1lnv2}^2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon}$ $+\frac{\text{lambdaH2lnv1}^2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$ $+\frac{5\text{lambdaH2lnv2}^2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon}$ $+\frac{\text{lambdaH1lnv1lambdaH1lnv2}2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon}$ $+\frac{\text{lambdaH2lnv1lambdaH2lnv2}2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon}$ $+\frac{1}{2}\text{lambdaH1lnv1}^2h\log\left(\frac{\mu\text{bar}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{5}{6}\text{lambdaH1lnv2}^2h\log\left(\frac{\mu\text{bar}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{\text{lambdaH1lnv1lambdaH1lnv2}2h\log\left(\frac{\mu\text{bar}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}}$ $+\frac{1}{2}\text{lambdaH2lnv1}^2h\log\left(\frac{\mu\text{bar}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{5}{6}\text{lambdaH2lnv2}^2h\log\left(\frac{\mu\text{bar}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{\text{lambdaH2lnv1lambdaH2lnv2}2h\log\left(\frac{\mu\text{bar}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}}$ $+\frac{h\overline{\lambda_{\tau3}}H_{a_1}H_{a_2}\lambda_{\tau3}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$ $+\frac{h\overline{\lambda_{\tau3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{\tau3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$ $+\frac{h\log\left(\frac{\mu\text{bar}^2}{M_{S_1}^2}\right)\overline{\lambda_{\tau3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{\tau3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$ $-\frac{h\overline{\lambda_{\tau3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{\tau3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$ $+\frac{h\log\left(\frac{\mu\text{bar}^2}{M_{S_2}^2}\right)\overline{\lambda_{\tau3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{\tau3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$	$-\frac{\hbar\left(W^{A_1\mu_2}\right)^2}{12\epsilon}$ $-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}^2}{M_{S_1}^2}\right)\left(W^{\mu_1\mu_2}\right)^2$ $-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}^2}{M_{S_2}^2}\right)\left(W^{\mu_1\mu_2}\right)^2$	$-\frac{h\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W_{1,1,1}W_{2,1,2}W_{1,3,1,2}^T}CP_L\ell_{a_2,1}H_{a_1}H_{a_4}M_F^2\lambda_{\tau1}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}M_F^2}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)}$ $+\frac{h\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W_{1,1,1}W_{2,1,2}W_{1,3,1,2}^T}CP_L\ell_{a_2,1}H_{a_1}H_{a_4}M_F^2\lambda_{\tau1}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}M_F^2}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)}$ $-\frac{h\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W_{1,1,1}W_{2,1,2}W_{1,3,1,2}^T}CP_L\ell_{a_2,1}H_{a_1}H_{a_4}M_F^2\lambda_{\tau1}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}M_F^2}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)}$ $+\frac{h\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W_{1,1,1}W_{2,1,2}W_{1,3,1,2}^T}CP_L\ell_{a_2,1}H_{a_1}H_{a_4}M_F^2\lambda_{\tau1}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}M_F^2}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)}$ $+\frac{h\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W_{1,1,1}W_{2,1,2}W_{1,3,1,2}^T}CP_L\ell_{a_2,1}H_{a_1}H_{a_4}M_F^2\lambda_{\tau1}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}M_F^2}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)}$ $+\frac{h\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W_{1,1,1}W_{2,1,2}W_{1,3,1,2}^T}CP_L\ell_{a_2,1}H_{a_1}H_{a_4}M_F^2\lambda_{\tau1}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}M_F^2}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)}$ $+\frac{h\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W_{1,1,1}W_{2,1,2}W_{1,3,1,2}^T}CP_L\ell_{a_2,1}H_{a_1}H_{a_4}M_F^2\lambda_{\tau1}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}M_F^2}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)}$ $+\frac{h\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W_{1,1,1}W_{2,1,2}W_{1,3,1,2}^T}CP_L\ell_{a_2,1}H_{a_1}H_{a_4}M_F^2\lambda_{\tau1}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}M_F^2}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)}$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$H \ell$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$ \begin{aligned} & -\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon} \\ & -\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)B_{\mu_1\mu_2}^2 \\ & -\frac{1}{4}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2 \end{aligned} $	$ \begin{aligned} & +C\lambda_{T3}H_{a_1}H_{a_1}^\dagger \\ & -H_{a_1}\mu^2H_{a_1}^\dagger \\ & +\frac{3\text{lambdaH1Inv}2^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ & +\frac{3\text{lambdaH2Inv}2^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ & +\frac{3}{2}\text{lambdaH1Inv}2^2\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger \\ & +\frac{3}{2}\text{lambdaH2Inv}2^2\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger \\ & +\frac{\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & +\frac{\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2} \\ & +\frac{\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2} \\ & -\frac{\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2} \\ & -\frac{\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2} \end{aligned} $	$ \begin{aligned} & -\frac{5\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{6\epsilon} \\ & -\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\ & -\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\ & -\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \end{aligned} $	$ \begin{aligned} & -\frac{i\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{y_{1,\tau_1,\tau_1}}y_{2,\tau_2,\tau_2}^*e_{a_1,\tau_2}^TC P_\tau e_{a_2,\tau_1}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}e_{a_1,\tau_2}e_{a_2,\tau_2}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)} \\ & +\frac{i\hbar\log\left(\frac{\mu_F^2}{M_{S_1}^2}\right)\overline{y_{1,\tau_1,\tau_1}}y_{2,\tau_2,\tau_2}^*e_{a_1,\tau_2}^TC P_\tau e_{a_2,\tau_1}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}e_{a_1,\tau_2}e_{a_2,\tau_2}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)} \\ & +\frac{i\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{\lambda_{T1}}e_{a_3,\tau_1}C P_R\overline{\lambda_{T4,\tau_4}}^*M_{S_1}^2y_{1,\tau_1,\tau_1}y_{2,\tau_2,\tau_2}^*e_{a_1,\tau_2}e_{a_3,\tau_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)} \\ & +\frac{i\hbar\log\left(\frac{\mu_F^2}{M_{S_1}^2}\right)\overline{\lambda_{T3}}e_{a_3,\tau_1}C P_R\overline{\lambda_{T4,\tau_4}}^*M_{S_1}^2y_{1,\tau_1,\tau_1}y_{2,\tau_2,\tau_2}^*e_{a_1,\tau_2}e_{a_3,\tau_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)} \\ & -\frac{i\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{y_{1,\tau_1,\tau_1}}y_{2,\tau_2,\tau_2}^*e_{a_1,\tau_2}^TC P_\tau e_{a_2,\tau_1}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}e_{a_1,\tau_2}e_{a_3,\tau_4}M_F}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)} \\ & -\frac{i\hbar\log\left(\frac{\mu_F^2}{M_{S_1}^2}\right)\overline{y_{1,\tau_1,\tau_1}}y_{2,\tau_2,\tau_2}^*e_{a_1,\tau_2}^TC P_\tau e_{a_2,\tau_1}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}e_{a_1,\tau_2}e_{a_3,\tau_4}M_F}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)} \\ & +\frac{i\hbar\log\left(\frac{\mu_F^2}{M_{S_1}^2}\right)\overline{\lambda_{T1}}e_{a_3,\tau_1}C P_R\overline{\lambda_{T4,\tau_4}}^*M_{S_1}^2M_{S_2}^2y_{1,\tau_1,\tau_1}y_{2,\tau_2,\tau_2}^*e_{a_1,\tau_2}e_{a_2,\tau_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)} \\ & +\frac{i\hbar\log\left(\frac{\mu_F^2}{M_{S_1}^2}\right)\overline{\lambda_{T3}}e_{a_3,\tau_1}C P_R\overline{\lambda_{T4,\tau_4}}^*M_{S_1}^2M_{S_2}^2y_{1,\tau_1,\tau_1}y_{2,\tau_2,\tau_2}^*e_{a_1,\tau_2}e_{a_2,\tau_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)} \\ & -\frac{i\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{\lambda_{T1}}e_{a_3,\tau_1}C P_R\overline{\lambda_{T4,\tau_4}}^*M_{S_1}^2M_{S_2}^2y_{1,\tau_1,\tau_1}y_{2,\tau_2,\tau_2}^*e_{a_1,\tau_2}e_{a_2,\tau_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)} \end{aligned} $

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$$\begin{aligned}
& \frac{3h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{16 (M_F^2 - M_{S_1}^2)^2} \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^4 \\
& - \frac{h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{4 (M_F^2 - M_{S_2}^2)^2} \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^4 \\
& - \frac{3h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{16 (M_F^2 - M_{S_1}^2)^2} \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^4 \\
& - \frac{h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{4 (M_F^2 - M_{S_2}^2)^2} \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^4 \\
& + \frac{3h y_{1,i_3,r_1} \ell_{a_1,i_2} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{8 (M_F^2 - M_{S_1}^2)} \\
& + \frac{3h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{8 (M_F^2 - M_{S_1}^2)} \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2 \\
& + \frac{3h y_{1,i_3,r_1} \ell_{a_1,i_2} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{16 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{3h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{8 (M_F^2 - M_{S_1}^2)^2} \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2 \\
& + \frac{h y_{2,i_3,r_1} \ell_{a_1,i_2} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{2 (M_F^2 - M_{S_2}^2)} \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2 \\
& + \frac{h y_{2,i_3,r_1} \ell_{a_1,i_2} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{2 (M_F^2 - M_{S_2}^2)^2} \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2 \\
& + \frac{3h y_{1,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c P_L \ell_{a_1,i_3} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{8 (M_F^2 - M_{S_1}^2)} \\
& + \frac{3h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{8 (M_F^2 - M_{S_1}^2)} \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2 \\
& + \frac{3h y_{1,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c P_L \ell_{a_1,i_3} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{16 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{3h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{8 (M_F^2 - M_{S_1}^2)^2} \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2 \\
& + \frac{h y_{2,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c P_L \ell_{a_1,i_3} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{2 (M_F^2 - M_{S_2}^2)} \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2 \\
& + \frac{h y_{2,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c P_L \ell_{a_1,i_3} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{4 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{h \log \left(\frac{\mu h \nu_2^2}{M_F^2} \right)}{2 (M_F^2 - M_{S_2}^2)^2} \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2 \\
& + \frac{3h y_{1,i_3,r_1} \ell_{a_1,i_2} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{16e} \\
& - \frac{3}{32} h y_{1,i_3,r_1} \ell_{a_1,i_2} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{3h y_{1,i_3,r_1} \ell_{a_1,i_2} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{8 (M_F^2 - M_{S_1}^2)} \\
& - \frac{3h \log \left(\frac{\mu h \nu_2^2}{M_{S_1}^2} \right)}{8 (M_F^2 - M_{S_1}^2)} \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{3h y_{1,i_3,r_1} \ell_{a_1,i_2} P_R e_{i_1} H_{a_1} M_{S_1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{16 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{h y_{2,i_2,r_1} \ell_{a_1,i_2} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4e} \\
& - \frac{1}{8} h y_{2,i_3,r_1} \ell_{a_1,i_2} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{h y_{2,i_3,r_1} \ell_{a_1,i_2} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{h \log \left(\frac{\mu h \nu_2^2}{M_{S_2}^2} \right)}{2 (M_F^2 - M_{S_2}^2)} \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{h y_{2,i_3,r_1} \ell_{a_1,i_2} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{h \log \left(\frac{\mu h \nu_2^2}{M_{S_2}^2} \right)}{4 (M_F^2 - M_{S_2}^2)^2} \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1} \\
& + \frac{3h y_{1,i_2,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c P_L \ell_{a_1,i_3} y_{1,i_3,r_1} H_{a_1}^\dagger}{16e} \\
& - \frac{3}{32} h y_{1,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c P_L \ell_{a_1,i_3} y_{1,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{3h y_{1,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c P_L \ell_{a_1,i_3} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{8 (M_F^2 - M_{S_1}^2)} \\
& + \frac{3h \log \left(\frac{\mu h \nu_2^2}{M_{S_1}^2} \right)}{8 (M_F^2 - M_{S_1}^2)} \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c} P_L \ell_{a_1,i_3} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{3h y_{1,i_1,r_1} Y_{e,i_3,i_2} \tilde{e}_{i_2}^c P_L \ell_{a_1,i_3} M_{S_1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{16 (M_F^2 - M_{S_1}^2)^2}
\end{aligned}$$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	e
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